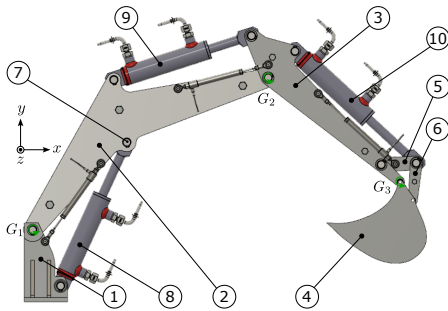
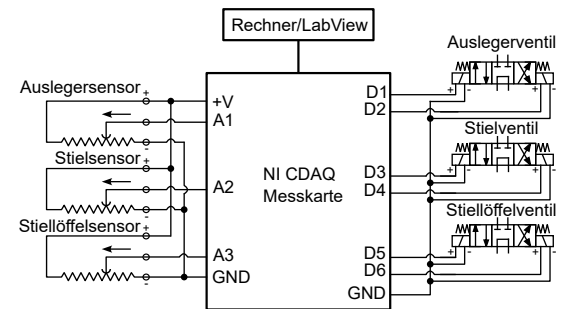


Bachelor's Thesis – Design and Construction of a Hydraulic Excavator Arm

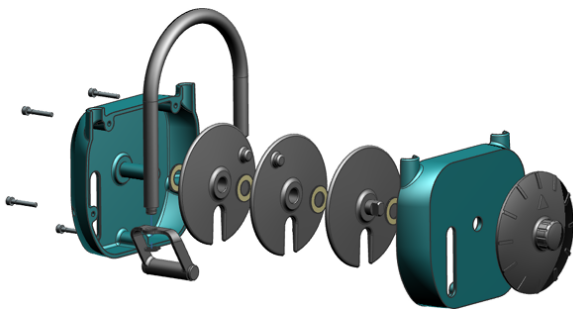


- CAD design of a hydraulic excavator arm for use in teaching
- Mechanical dimensioning and component selection
- Integration of hydraulic and electrical subsystems

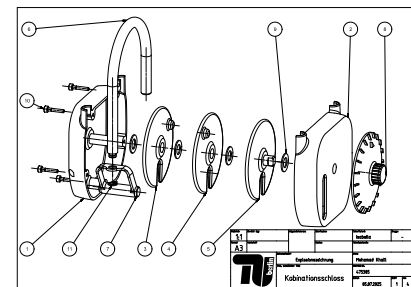


- Development and documentation of the electrical system architecture
- Integration of sensors and an NI-CDAQ data acquisition device
- Control of electrically actuated 4/3-way valves

Design of a Combination Lock in Siemens NX



- Modeling of the individual components of a combination lock in Siemens NX
- Parametrization of screws and holes using expressions
- Creation of a fully constrained assembly with assigned materials

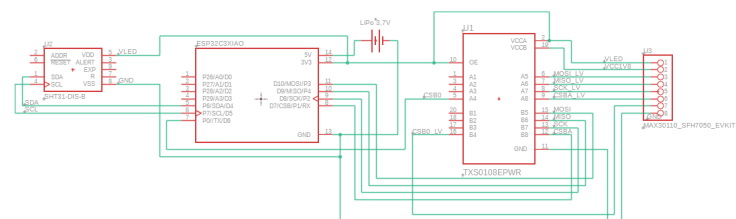


- Creation of an exploded view of the complete assembly
- Derivation of a bill of materials, assembly drawing, and fit table
- Technical documentation according to specified drawing and submission guidelines

Development of a Sensor System for Differentiated Vital Parameter Measurement



- Development of a compact sensor system for measuring vital parameters
- Selection and integration of the MAX30102 sensor for measuring heart rate and oxygen saturation
- Integration of the SHT31 sensor for measuring temperature and relative humidity



- Overall electrical system design with the XIAO ESP32C6 as the central control unit
- Sensor communication via I2C and display connection via SPI/I2C
- Design of the power supply for mobile, battery-powered operation